ModelIt!

Student Activity Pack

Should We or Shouldn't We?

When Science Meets Hard Choices

Grade NE | NGSS HS-ETS1-1, HS-ETS1-3 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever had to make a decision where there was no clearly right answer? What did you do?**

2. **Should scientists always use new technology to fix environmental problems? Why or why not?**

3. **What does 'unintended consequences' mean? Can you think of an example?**

4. **Is it ever wrong to help — even if your intentions are good?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Problem Severity | Solution Risk | Available Evidence | Intervention Justified

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Clear Yes

Severe problem, safe solution, strong evidence

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Clear No

Minor problem, risky solution, weak evidence

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Hard Call

Severe problem, risky solution, moderate evidence

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Cost — How expensive the intervention is — limited budgets mean not everything can be funded

Public Support — Whether the community supports or opposes the intervention

Reversibility — Whether the intervention can be undone if it goes wrong

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do we decide whether to use technology to fix an environmental problem — even if the fix might cause new problems? Use evidence from your simulation to support your answers.

1. **Why is it important to consider ALL three factors (severity, risk, evidence) before making a decision?**

2. **In the 'Hard Call' scenario, what additional information would change your decision?**

3. **Can a model ever tell you what is morally right? What can it tell you?**

4. **Think of a real-world example where good intentions led to unintended consequences. What went wrong?**

5. **How does the amount of evidence change how much risk you are willing to accept?**

8 STEM Challenge: Ethics Courtroom

ModelIt!

YOUR ENGINEERING MISSION

In structured debates, argue FOR or AGAINST technological intervention in real conservation scenarios. Use your model data as evidence.

THE SCENARIO

A conservation agency must decide whether to use new technology to help a struggling ecosystem. Two expert teams will present their cases — one arguing for intervention, one against. A jury of classmates will decide based on the evidence.

Engineering Design Thinking

1. HOW WILL YOU RATE PROBLEM SEVERITY ON A SCALE OF 1-10 WITH EVIDENCE?
2. WHAT SPECIFIC RISKS HAVE BEEN DOCUMENTED FOR YOUR PROPOSED SOLUTION?
3. WHAT DOES THE PEER-REVIEWED RESEARCH SAY ABOUT THIS APPROACH?
4. HOW WILL YOU ADDRESS THE STRONGEST COUNTERARGUMENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental Ethicists and Science Policy Advisors help governments and organizations make difficult decisions about technology, conservation, and environmental intervention. They earn $65,000–$120,000/year.